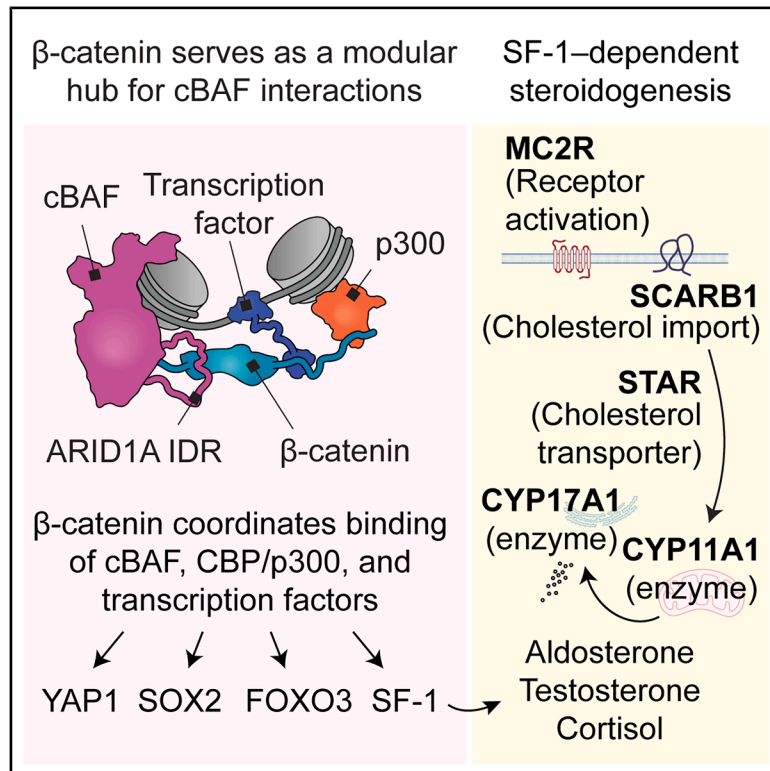


β-catenin functions as a molecular adapter for disordered cBAF interactions

Graphical abstract



Authors

Yuen San Chan, Qinyu Gao, Sarah A. Robinson, ..., Vaclav Veverka, Katerina Cermakova, H. Courtney Hodges

Correspondence

katerina.cermakova@bcm.edu (K.C.), chodges@bcm.edu (H.C.H.)

In brief

By dissecting steroidogenic expression dependencies, Chan et al. reveal that β-catenin mediates physical association of cBAF with key binding partners. The authors show that a disordered region of cBAF subunit ARID1A interacts with Armadillo repeats on β-catenin, highlighting roles for modular adapters in IDR-mediated interactions.

Highlights

- Dissection of functional interactions involving β-catenin and cBAF (SWI/SNF)
- Disordered sequences of ARID1A directly bind folded Armadillo repeats on β-catenin
- β-catenin mediates cBAF interaction with SF-1 (NR5A1), YAP1, SOX2, FOXO3, and CBP/p300
- Contact with structured protein mediates key IDR-driven cBAF interactions



Article

β -catenin functions as a molecular adapter for disordered cBAF interactions

Yuen San Chan,^{1,2} Qinyu Gao,^{1,2} Sarah A. Robinson,^{1,2} Wenzhi Wang,^{1,2} Ruzena Filandrova,³ Lisa-Maria Weinhold,^{3,4} Mario Loeza Cabrera,^{1,2} Miao Zhang,⁵ Chandra Shekar R. Ambati,⁶ Antonio M. Lerario,⁷ Nagireddy Putluri,^{1,6,8} Katja Kiseljick-Vassiliades,^{9,10} Margaret E. Wierman,^{9,10} Mouhammed Amir Habra,¹¹ Gary D. Hammer,^{7,12} Vaclav Veverka,^{3,4} Katerina Cermakova,^{2,13,*} and H. Courtney Hodges^{1,2,8,14,15,*}

¹Department of Molecular and Cellular Biology, Baylor College of Medicine, Houston, TX 77030, USA

²Center for Precision Environmental Health, Baylor College of Medicine, Houston, TX 77030, USA

³Institute of Organic Chemistry and Biochemistry, Czech Academy of Sciences, Prague, Czech Republic

⁴Department of Cell Biology, Faculty of Science, Charles University, Prague, Czech Republic

⁵Department of Pathology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁶Advanced Technology Cores, Baylor College of Medicine, Houston, TX 77030, USA

⁷Department of Internal Medicine, Division of Metabolism, Endocrinology and Diabetes, University of Michigan, Ann Arbor, MI 48105, USA

⁸Dan L Duncan Comprehensive Cancer Center, Baylor College of Medicine, Houston, TX 77030, USA

⁹Division of Endocrinology, Metabolism and Diabetes, Department of Medicine, University of Colorado School of Medicine at Colorado Anschutz Medical Campus, Aurora, CO 80045, USA

¹⁰Research Service Rocky Mountain Regional Veterans Affairs Medical Center, Aurora, CO 80045, USA

¹¹Department of Endocrine Neoplasia and Hormonal Disorders, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

¹²Endocrine Oncology Program, Rogel Cancer Center, University of Michigan Health System, Ann Arbor, MI 48109, USA

¹³Department of Biochemistry and Molecular Pharmacology, Baylor College of Medicine, Houston, TX 77030, USA

¹⁴Department of Bioengineering, Rice University, Houston, TX 77030, USA

¹⁵Lead contact

*Correspondence: katerina.cermakova@bcm.edu (K.C.), chodges@bcm.edu (H.C.H.)

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SUMMARY

BAF (SWI/SNF) chromatin remodelers engage binding partners to generate site-specific DNA accessibility. However, the basis for interaction between BAF and divergent binding partners has remained unclear. Here, we tested the hypothesis that scaffold proteins augment BAF's binding repertoire by examining β -catenin (CTNNB1) and steroidogenic factor 1 (SF-1, NR5A1), a transcription factor central to steroid production in human cells. BAF inhibition rapidly opposed SF-1/ β -catenin enhancer occupancy, impairing SF-1 target activation and SF-1/ β -catenin autoregulation. These effects arise due to β -catenin's role as a molecular adapter between SF-1 and an intrinsically disordered region (IDR) of the canonical BAF (cBAF) subunit ARID1A. In contrast to exclusively IDR-driven mechanisms, adapter function is mediated by direct association of ARID1A with β -catenin's folded Armadillo repeats. β -catenin similarly linked cBAF to YAP1, SOX2, FOXO3, and CBP/p300, reflecting a general IDR-mediated mechanism for modular coordination between factors. Molecular visualization highlights β -catenin's adapter role for interaction of cBAF with binding partners.

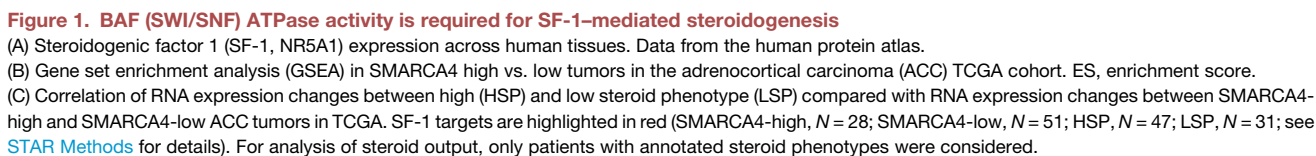
INTRODUCTION

BAF (SWI/SNF) family ATP-dependent chromatin remodelers act as critical co-activators in many settings. The coupling of catalytic ATPase activity with non-catalytic interactions involving BAF complexes is essential to enhancer function,^{1–3} opposition to constitutive and facultative heterochromatin,^{4,5} replication,⁶ response to DNA damage,^{7,8} transcriptional pausing,⁹ and many other activities.^{10,11} The high degree of cell plasticity enabled by these processes arises in part from the diverse inter-

actomes of BAF complexes,^{12,13} which include selective interactions with transcription factors (TFs),¹⁴ histone-modifying enzymes,¹⁵ proteins that mediate chromatin architecture,¹⁶ transcription elongation,¹⁷ RNA splicing,⁹ DNA damage,⁷ and many other cofactors.^{12,13,18,19} Collectively, these interactions give rise to highly tissue- and state-specific functions.

The core architectures of BAF family complexes have revealed the structural basis for their capacity to mobilize nucleosomes.^{20–23} BAF chromatin remodelers assemble the primary ATPase subunit SMARCA4 (BRG1) into at least three major





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sub-complexes that are defined by their unique subunit compositions.²⁴ For example, ARID1A is found within canonical (cBAF) complexes,²⁵ whereas PBRM1 and BRD7 are found within polybromo-associated (PBAF) complexes,²⁶ and BRD9 is found in non-canonical (ncBAF or GBAF) complexes.^{24,27} Instead of the well-ordered cores, which power nucleosome mobilization, intrinsically disordered regions (IDRs) have been attributed as the primary interaction surfaces that mediate these complexes' capacity to engage diverse interaction partners.^{13,28,29} Unfortunately, the IDRs on these complexes are highly dynamic and remain structurally unresolved. As a result, many questions have remained regarding the biochemical basis for the extensive interactions of BAF complexes with TFs and other chromatin machinery.

Disordered protein has emerged as an important mediator of protein interactions, particularly for nuclear proteins involved in chromatin and transcription.²⁸ IDRs have a high capacity to engage other IDRs through heterotypic interactions^{13,30–33}; however, the biochemical mechanisms that enable selectivity in IDR-IDR interactions remain incompletely understood. By contrast, a growing number of selective interactions have been identified whereby IDRs engage well-ordered folded protein domains with high specificity.^{28,34–36} IDR-domain affinities are highly variable but substantially overlap with those observed between folded proteins³⁷ and are sufficient to sustain protein-protein interactions.^{34,38–40}

We hypothesized that folded domains with more extensive interaction surfaces could mediate selective interactions with IDRs in BAF complexes and thereby expand the BAF interaction repertoire in a modular fashion. We therefore sought to identify a candidate protein that interacts both with TFs and BAF complexes and whose proximity is essential for transcription activation. These requirements were readily fit by β -catenin (CTNNB1), a protein primarily composed of folded Armadillo repeats and IDRs that plays scaffolding roles in other settings.⁴¹ β -catenin interacts with TCF/LEF and BAF complexes during canonical Wnt signaling^{42–44} but also has selective interactions with multiple TFs and co-activators, including those from YAP,^{45,46} SOX,^{45,46} and FOXO⁴⁷ families, as well as the nuclear receptor TF known as steroidogenic factor 1 (SF-1, NR5A1).^{48,49} β -catenin also directly interacts with histone acetyltransferases CBP/p300,³⁹ which cooperate with BAF complexes to regulate TF activity and gene expression.^{50,51} The overlap with BAF's binding partners as well as the lack of a known DNA-binding domain and the extensive interaction surface of β -catenin prompted us to hypothesize that it may serve as a molecular adapter for interactions between BAF complexes and shared interaction partners.

Compared with many other TFs, SF-1 activates a relatively consistent set of target genes.⁵² Ectopic expression of SF-1 in many settings is sufficient to activate the entire steroid biosynthesis pathway, including the cholesterol importer STAR and cytochrome P450 enzymes like CYP17A1 that are essential for steroid biosynthesis.^{53–55} As a result, SF-1 expression is generally restricted to endocrine tissues (Figure 1A). The highly defined transcriptional effects of SF-1 across many cell types make SF-1 an ideal TF to test and dissect the potential adapter role of β -catenin for BAF complexes and to evaluate currently unknown roles of BAF's involvement in steroid hormone production.

Here, we evaluated a role for β -catenin as a molecular adapter for chromatin remodeling by dissecting BAF-related functions in SF-1-mediated steroidogenesis. Our findings reveal that cBAF remodeling activity enables SF-1/ β -catenin binding at enhancers to activate steroidogenic gene expression, as well as to sustain autoregulation of the steroidogenic circuitry. We find that β -catenin acts as an essential molecular adapter that mediates binding of cBAF not only to SF-1 but also to an array of other β -catenin binding partners, including diverse TFs and CBP/p300. By dissecting the specific protein elements needed for biochemical association between the involved factors, we determined that an IDR in the cBAF-specific subunit ARID1A mediates direct physical association with folded Armadillo repeats on β -catenin, highlighting key roles for stably folded molecular adapters in orchestrating interactions of disordered regions on cBAF with TFs and chromatin regulators.

RESULTS

Excess SF-1-dependent hormone production is linked with high SMARCA4 and β -catenin

To interrogate BAF- and β -catenin-related mechanisms for SF-1 activity, we identified adrenocortical carcinoma (ACC), a tumor with highly activated SF-1/ β -catenin transcriptional programs and a high frequency of excess steroid hormone production,^{48,56} as an ideal experimental setting. Analysis of ACC data from The Cancer Genome Atlas (TCGA; Figures S1A–S1F)⁵⁶ revealed that ACC tumors with high SMARCA4 expression exhibit higher expression of SF-1 targets (gene set enrichment analysis [GSEA] normalized enrichment score [NES] = 2.0, p = 2.9e–6), steroidogenic genes (NES = 1.6, p = 0.03), and β -catenin targets (NES = 1.8, p = 7.2e–4) compared with tumors with low SMARCA4 expression (Figure 1B; Table S1). Moreover, global gene expression patterns between these two groups strongly correlated with expression changes between high (HSP) and low steroid phenotype (LSP) groups (R = 0.76, p < 2.2e–16,

(D) Representative IHC images of steroidogenic expression signature (SES) components SMARCA4, β -catenin, and SF-1 in ACC by steroid phenotype.

(E) Heatmap of unbiased hierarchical clustering of validation cohort of ACC tumors. Tumors are assigned to HSP or LSP based on clinical detection of excess steroid hormone production. Pie charts indicate the proportions of tumors with HSP in each category.

(F) Expression changes induced by BRM014 in H295R cells, as measured by RNA-seq. SF-1 targets are presented in red, and β -catenin targets are presented in blue.

(G) Gene sets from differentially reduced transcripts upon BRM014 treatment in H295R cells ranked by normalized enrichment score (NES).

(H) BRM014-induced expression changes (filled circles) from RNA-seq presented in (F). Inset bar charts indicate DMSO-normalized production of individual steroid hormones or precursors in H295R cells, as assessed by ELISA or LC-MS. Error bars are mean $\pm$ SEM (n = 3 independent replicates).

* p < 0.05, ** p < 0.01, **** p < 2.2e–16.

See also Figures S1–S3 and Table S1.

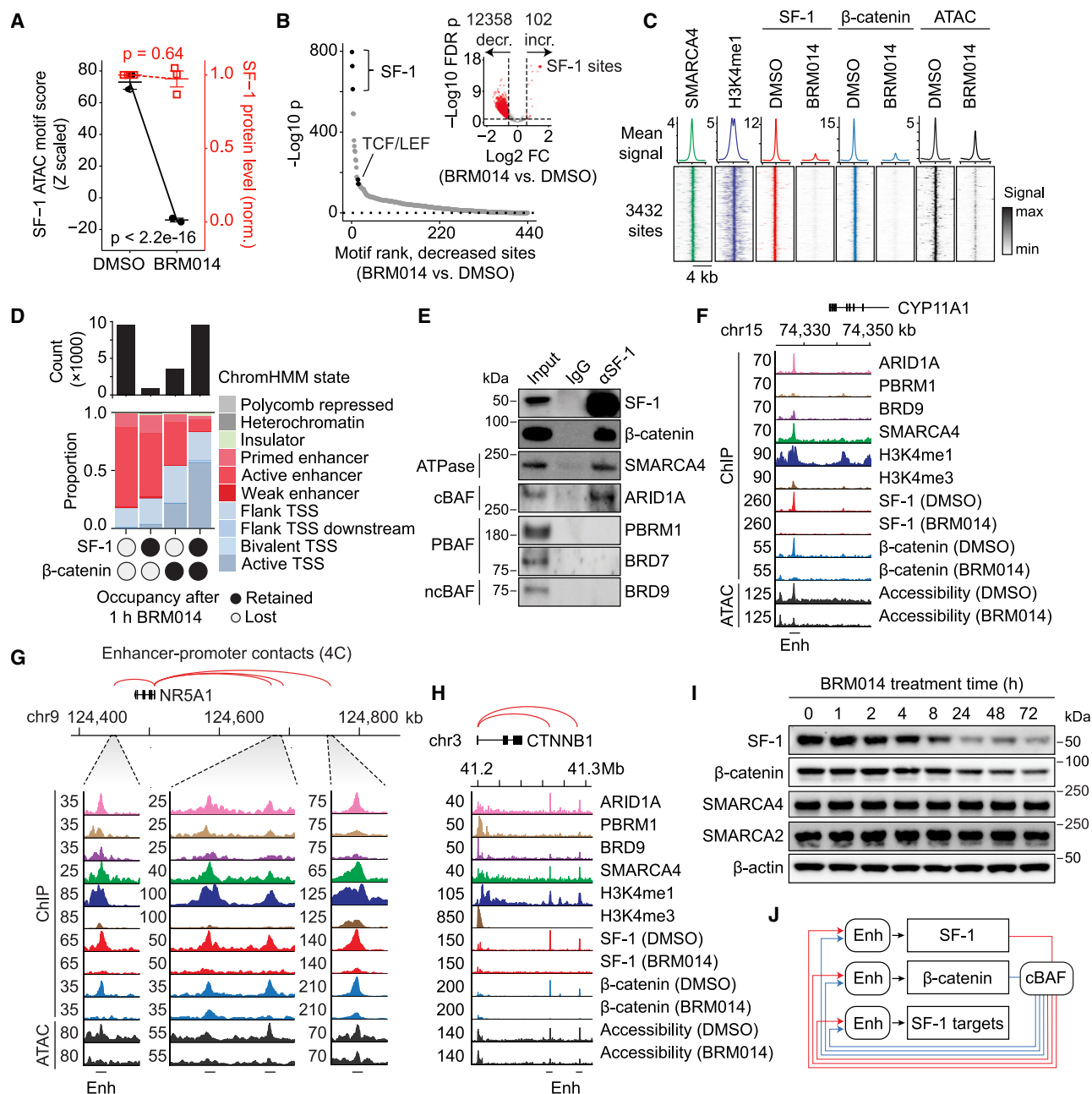


Figure 2. cBAF mediates DNA accessibility and SF-1/ β -catenin binding at enhancers to sustain autoregulatory steroidogenic expression
(A) SF-1 motif DNA accessibility (black) and SF-1 protein levels (red) measured after 1 h treatment with BRM014 or DMSO control. Error bars are mean $\pm$ SEM (ATAC-seq $n = 2$ independent replicates, protein levels $n = 3$ independent replicates). Protein levels were measured using western blot densitometry. p value for ATAC-seq from hypergeometric test and p value for protein levels from t test.
(B) Ranking of TF motif enrichment based on p value from sites that lose accessibility. Volcano plot (inset) shows differential DNA accessibility in H295R cells treated with 1 h of BRM014 vs. DMSO control.
(C) Heatmap of genomic SMARCA4-regulated sites that show reduced SF-1 and β -catenin occupancy along with reduced DNA accessibility following 1 h of BRM014 treatment.
(D) Classification of individual clusters with differential loss of SF-1/ β -catenin at different genomic regions defined by ChromHMM using histone marks. Bar plot indicates the number of sites in each cluster.
(E) Co-immunoprecipitation (CoIP) of endogenous proteins using anti-SF-1 antibodies in H295R cells ($n = 2$ independent replicates).
(F) Browser tracks of SF-1 target CYP11A1 locus showing loss of SF-1/ β -catenin and reduced DNA accessibility at cBAF-regulated enhancer (Enh) following 1 h of BRM014 treatment or DMSO control.

(legend continued on next page)

Figure 1C; Table S1). No association was observed for the minor BAF ATPase SMARCA2 (BRM, Figures S1C–S1F), underscoring the unique relationship between SF-1, β -catenin, and SMARCA4. Unsupervised clustering of tumors based on expression levels of SF-1 (NR5A1), β -catenin (CTNNB1), and SMARCA4 enabled classification based on their joint expression, which we termed the steroidogenic expression signature (SES). Elevated RNA levels of this signature (SES RNA-high) were highly predictive of HSP (chi-square test, $p = 1.7e-5$; Figures S1G and S1H). Compared with SF-1 and β -catenin RNA levels without SMARCA4, the addition of SMARCA4 levels significantly improved classification accuracy for HSP (ANOVA, $p < 0.03$). Hence, elevated expression of steroidogenic pathways *in vivo* is associated with high RNA levels of SF-1, β -catenin, and SMARCA4.

We performed immunohistochemistry (IHC) of a tissue microarray composed of an independent cohort of primary human ACC tumors ($N = 35$), adrenal adenomas (ACA, $N = 11$), and normal adrenal cortex tissue samples ($N = 11$; Figures S2A–S2D). In non-malignant ACA, neither β -catenin nor SMARCA4 was overexpressed ($p > 0.05$). However, SMARCA4 ($p = 2.7e-8$) and β -catenin ($p = 2.7e-3$) were both overexpressed at the protein level in malignant ACC (Figures S2C and S2D). Similarly, 100% of SES protein-high tumors overproduced steroid hormones (Figures 1D, 1E, and S2E–S2G; chi-square test, $p < 0.01$). Our analyses link high SMARCA4 and β -catenin protein levels to SF-1-dependent hormone production in humans.

SMARCA4 regulates SF-1 transcriptional activity

To test whether BAF catalytic activity is required for SF-1-dependent gene expression, we treated two human steroidogenic ACC cell lines, H295R⁵⁷ and CU-ACC1,⁵⁸ with the SMARCA4/2 ATPase inhibitor BRM014^{17,59–61} for 72 h and performed RNA sequencing (RNA-seq). BAF ATPase inhibition strongly reduced the expression of SF-1 targets that play key roles in steroid metabolism, including the adrenocorticotrophic hormone (ACTH) receptor MC2R, cholesterol receptor SCARB1, cholesterol transporter STAR, and cytochrome P450 enzymes CYP11A1 and CYP17A1, as well as β -catenin targets (Figures 1F and S3A; Table S1). GSEA revealed that SF-1 targets (NES = -2.9 , $p = 6.3e-27$) and genes associated with steroidogenesis (NES = -1.7 , $p = 0.02$) were among the most significantly downregulated gene sets (Figures 1G, S3B, and S3C; Table S1). These effects were reproducible upon treatment with the orthogonal SMARCA4/2 ATPase inhibitor FHD-286 or degrader AU-15330⁶² (Figures S3D and S3E), confirming on-target effects. Consistent with these changes, the levels of hormones aldosterone (t test $p = 6.1e-4$), cortisol ($p = 4.4e-3$), and testosterone ($p = 9.4e-3$), as well as their precursors, were significantly decreased upon BAF inhibition, as assessed by ELISA and liquid chromatography-mass spec-

trometry (LC-MS) (Figures 1H and S3F). Our results demonstrate that BAF ATPase activity is essential for steroid hormone production in steroid-producing cells.

Consistent with the human data, short hairpin RNA (shRNA) knockdown revealed that SF-1 targets primarily depend on SMARCA4 for their expression, while only a subset depends on the alternative ATPase SMARCA2 (Figures S3G and S3H). Our findings establish SMARCA4 as the primary BAF ATPase governing the SF-1 steroidogenic program.

cBAF enables enhancer binding of SF-1 and β -catenin

To determine how BAF ATPase activity influences the chromatin occupancy of SF-1, we began by measuring changes in DNA accessibility at sites bearing SF-1 motifs following acute inhibition using BRM014 in H295R cells. We observed pronounced genome-wide losses of DNA accessibility at sites matching SF-1 consensus motifs within 1 h of BRM014 treatment ($p < 2.2e-16$), a time when SF-1 protein levels were unchanged ($p = 0.64$; Figure 2A). While other motifs associated with β -catenin binding partners, such as TCF/LEF, had reduced accessibility, reductions at sites matching the SF-1 motif were the most significantly associated with decreased accessibility (Figure 2B; Table S2).

Chromatin profiling revealed strongly overlapping genome-wide binding patterns of SF-1, β -catenin, and SMARCA4 (Figure S4A), with sites overlapping SMARCA4 having the strongest occupancy of SF-1 and β -catenin (Figure S4B). We furthermore identified 3,432 genomic sites jointly bound by SMARCA4, SF-1, and β -catenin that showed reduced DNA accessibility along with reduced binding of SF-1 and β -catenin following 1 h of BRM014 treatment (Figures 2C and S4C). These sites were also highly enriched in the enhancer mark H3K4me1 and depleted of the active promoter mark H3K4me3 (Figure S4D), consistent with reduced dependency on BAF at promoters.^{17,60,61} These sites were, moreover, associated with loss of CBP/p300 binding (Figure S4E), indicating coordination between BAF and histone acetyltransferase activity. Annotation of genome-wide positions using a 10-state ChromHMM⁶³ classifier confirmed that sites bearing losses of SF-1 and β -catenin were enriched at enhancers and depleted at transcription start sites (Figures 2D and S4F–S4H).

Immunoprecipitation (IP) of SF-1 confirmed its physical interaction with β -catenin⁴⁸ and revealed its interaction with cBAF subunits ARID1A and SMARCA4 (Figure 2E). By contrast, no interaction was detected with PBAF subunits PBRM1 or BRD7, or with ncBAF subunit BRD9 (Figure 2E). In agreement, BAF inhibition induced joint reduction of SF-1 and β -catenin binding at enhancers of SF-1 target genes associated with high occupancy of cBAF subunit ARID1A and low levels of PBAF subunit PBRM1 or ncBAF subunit BRD9 (Figures 2F, S4D, and S4F). These positions were also significantly enriched near SF-1 target

(G and H) (G) Browser tracks of SF-1 (NR5A1) and (H) β -catenin (CTNNB1) loci. Enhancers (Enh) identified by 4C anchored to the SF-1 promoter or previously identified for β -catenin are indicated in red. Identities of browser track rows for (G) are the same as in (H).

(I) Time course western blot of SF-1, β -catenin, SMARCA4, and SMARCA2 during 1 μ M BRM014 treatment in H295R cells ($n = 3$ independent replicates for 1 and 72 h).

(J) Schematic of the autoregulatory steroidogenic loop sustained by cBAF activity. Red lines indicate SF-1 activity, and blue lines indicate β -catenin activity. See also Figures S4 and S5 and Table S2.

genes whose expression levels were reduced upon BAF inhibition ($p = 2.7 \times 10^{-13}$; [Figures S4I–S4K](#)). Despite earlier work suggesting a unique role of PBAF for nuclear receptor activity,⁶⁴ our results demonstrate that cBAF complexes act as direct co-activators of SF-1 at enhancers. We conclude that physical interaction of cBAF complexes with SF-1 and β -catenin generates chromatin accessibility for their binding at enhancers involved in steroidogenesis.

cBAF sustains feedforward expression of SF-1 and β -catenin

cBAF-regulated SF-1/ β -catenin sites included enhancers that make long-distance contact with the promoters of NR5A1 (encoding SF-1) and CTNNB1 (β -catenin; [Figures 2G and 2H](#)). We therefore hypothesized that cBAF complexes are required to maintain the autoregulatory expression loop of SF-1 and β -catenin. We extended BRM014 treatment and confirmed that the protein levels of both SF-1 and β -catenin progressively diminished following sustained BAF inhibition without altering the expression of BAF ATPases SMARCA4 and SMARCA2 ([Figures 2I and S5A](#)). Additionally, shRNA knockdown confirmed that expression of SF-1 and β -catenin was mutually dependent ([Figures S5B and S5C](#)) and that SMARCA4 activity is upstream of both SF-1 and β -catenin ([Figure S3G](#)). Our results demonstrate that cBAF sustains feedforward autoregulatory expression of SF-1 and β -catenin by maintaining their binding at their own enhancers ([Figure 2J](#)), thereby representing an essential component of the self-sustaining expression circuit observed in steroid-producing tissues.^{48,65}

β -catenin is a molecular adapter between cBAF and SF-1

To dissect functional interactions in a context where SF-1 and β -catenin are not genetically essential or subject to confounding autoregulation, we used HEK 293T cells with ectopic expression of SF-1 at physiological levels ([Figures 3A and S6A](#)). Compared with empty vector control, SF-1 expression induced strong upregulation of canonical SF-1 targets CYP17A1 and STAR ([Figure 3B](#)), which served as markers of SF-1 activity and provided a setting for biochemical dissection. Knockout (KO) of β -catenin revealed that it was required for SF-1-dependent upregulation of SF-1 targets ([Figures 3A and 3B](#)) and for physical interaction between BAF complexes and SF-1 ([Figure 3C](#)). By contrast, physical interaction of cBAF and β -catenin was independent of SF-1 ([Figure S6B](#)). This led us to hypothesize that β -catenin acts as a molecular adapter between cBAF and SF-1. To dissect the protein elements responsible for physical association with SF-1 and cBAF complexes, we generated a set of hemagglutinin (HA)-tagged β -catenin truncation and deletion mutants ([Figure 3D](#)) and expressed them in HEK 293T cells. IP of these constructs revealed that β -catenin's C-terminal IDR is necessary for binding SF-1 ([Figures 3E and 3F](#)). Using a similar approach, we demonstrated the SF-1 ligand-binding domain (LBD) was required for physical association between SF-1 and β -catenin ([Figures 3G, 3H, and S6C](#)). Our results in human cells agree with previous reports that a region in the LBD is essential for murine SF-1 to bind to β -catenin in yeast two-hybrid studies.⁶⁶ By contrast, all of β -catenin's disordered regions, including the C-terminal IDR, were dispensable for interaction

with BAF complexes via SMARCA4 IP, which was instead lost following deletion of β -catenin's well-ordered Armadillo repeats 4–6 ([Figures 3I and 3J](#)). IP of ARID1A confirmed complete loss of interaction between β -catenin and cBAF complexes upon deletion of Armadillo repeats 4–6 ([Figure 3K](#)). Impaired contact between β -catenin and either cBAF or SF-1 was associated with failure to fully activate SF-1 transcription activity ([Figure S6D](#)). Our studies confirm that β -catenin acts as a molecular adapter between cBAF and SF-1 ([Figure 3L](#)).

IDR2 of ARID1A mediates association of cBAF with β -catenin

BRM014 treatment ([Figures 4A and 4B](#)) or dual knockout of cBAF subunits ARID1A and ARID1B ([Figures 4C, 4D, S7A, and S7B](#)) abolished the activation of SF-1 targets in HEK 293T cells. In contrast to ARID1A/B, knockout of cBAF subunits DPF1, DPF2, or DPF3, PBAF subunit PBRM1, or ncBAF subunit BRD9 did not abolish SF-1 transcription activity ([Figures S7C–S7F](#)). Among non-catalytic subunits, these experiments confirm a unique role for cBAF's ARID subunit in the activation of SF-1 targets.

To identify a minimal protein element of ARID1A that mediates SF-1 transcriptional activity, we designed full-length (FL) or sequential deletion mutants of FLAG-tagged ARID1A ([Figure 4E](#)). Each of these constructs contains the core binding region (CBR) to ensure complex integrity and inclusion of downstream subunits during cBAF assembly.²⁴ We expressed these constructs in dual ARID1A/ARID1B knockout HEK 293T cells expressing SF-1 ([Figure S8A](#)). Using STAR as a reporter of SF-1 activity revealed that, compared with FL constructs, the N-terminal IDR (IDR1) and ARID domain were dispensable, but deletion of ARID1A's second IDR (IDR2) strongly impaired SF-1 activity ([Figure 4F](#)). Loss of IDR2 furthermore uniquely eliminated cBAF binding to β -catenin as observed using IP of either SMARCA4 ([Figure 4G](#)) or ARID1A ([Figure S8B](#)), while preserving cBAF complex integrity as evidenced by inclusion of downstream cBAF subunit DPF2 ([Figures 4G and S8B](#)). Targeted rather than sequential deletions ([Figures S8C and S8D](#)) revealed a degree of functional redundancy of the IDRs. However, the sufficiency of IDR2 to mediate cBAF binding to β -catenin ([Figures 4G and S8B](#)) defined it as a minimal region enabling this interaction.

The ARID1A IDR2 directly interacts with β -catenin Armadillo repeats

The above results using cell-based assays raised two key questions: (1) does loss of β -catenin Armadillo repeats 4–6 induce only local structural changes, or does it impact the long-range organization of the Armadillo domain? (2) Which regions of ARID1A IDR2 and the Armadillo repeats, if any, directly interact? To answer these questions, we expressed and purified recombinant ARID1A IDR2, as well as the β -catenin Armadillo repeat domain with or without repeats 4–6 ([Figures 5A and S9A](#)). We then performed cross-linking mass spectrometry (XL-MS; [Figure 5B](#)) using these constructs. Deletion of repeats 4–6 within the Armadillo domain led to significantly shorter homotypic cross-links detected between repeats (t test $p = 0.0002$; [Figure 5C](#); [Table S3](#)). Pairwise contact plots revealed that

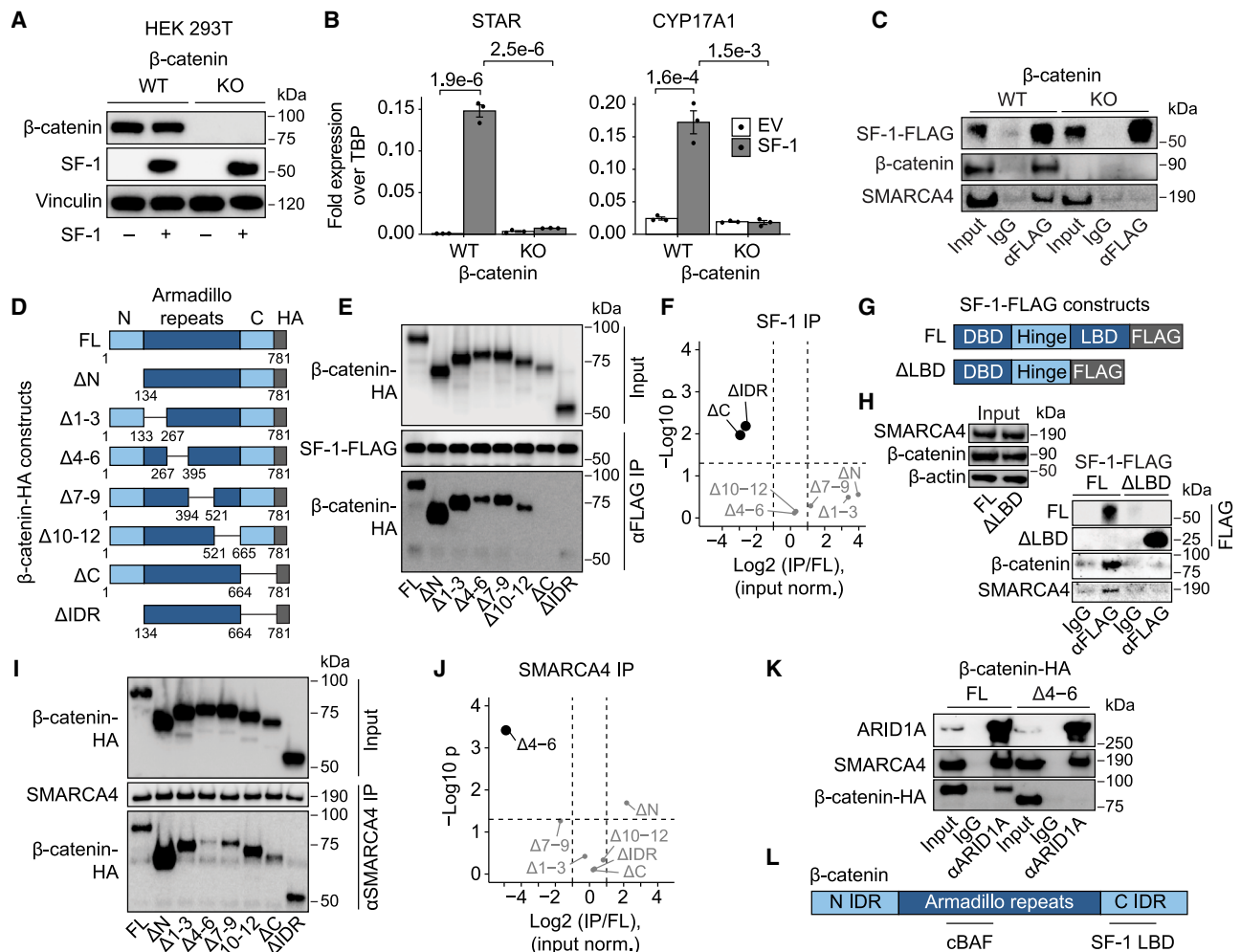


Figure 3. β-catenin functions as an adapter for cBAF interaction with SF-1

(A) Western blot of wild-type (WT) or knockout (KO) of β-catenin with ectopic expression of SF-1 (+) or empty vector (EV) control (–) in HEK 293T cells.
 (B) qPCR analysis of SF-1 targets STAR and CYP17A1 in HEK 293T cells with WT or KO β-catenin and EV or SF-1. Error bars are mean ± SEM ($n = 3$ independent replicates).
 (C) CoIP with SF-1-FLAG in HEK 293T cells expressing WT or KO β-catenin ($n = 2$ independent replicates).
 (D) Design of HA-tagged β-catenin deletion mutants.
 (E) CoIP of β-catenin deletion constructs with FLAG-tagged SF-1 ($n = 3$ independent replicates).
 (F) Volcano plot of co-immunoprecipitation densitometry in (E) ($n = 3$ independent replicates). Constructs without significant loss of interaction with SF-1 are shown in gray.
 (G) Design of FLAG-tagged SF-1 constructs. Full length (FL) and deletion of the ligand-binding domain (ΔLBD) are indicated.
 (H) CoIP of SMARCA4, β-catenin, and SF-1 in HEK 293T cells expressing either FL or ΔLBD SF-1, using immunoglobulin G (IgG) control or anti-FLAG antibodies. Input samples are shown separately (inset).
 (I) CoIP of β-catenin deletion constructs with SMARCA4 using SMARCA4 antibody ($n = 3$ independent replicates).
 (J) Volcano plot of co-immunoprecipitation densitometry in (I) ($n = 3$ independent replicates). Constructs without significant loss of interaction with SMARCA4 are shown in gray.
 (K) Validation of altered cBAF interaction by co-immunoprecipitation of FL or Δ4–6 β-catenin using IgG control or anti-ARID1A antibody.
 (L) Schematic of interactions between cBAF, β-catenin, and SF-1.
 Abbreviations are as follows: IDR, intrinsically disordered region; DBD, DNA-binding domain; LBD, ligand-binding domain.
 See also Figure S6.

deletion of repeats 4–6 eliminated long-range cross-links involving other repeats (Figure 5D). Additionally, the only homotypic cross-links observed in both the intact domain and following deletion of repeats 4–6 were those found entirely within

repeats 1–3 (Figure 5D). Hence our XL-MS data are consistent with disrupted structural organization across the Armadillo domain. Importantly, we also detected multiple heterotypic cross-links between ARID1A IDR2 and β-catenin Armadillo

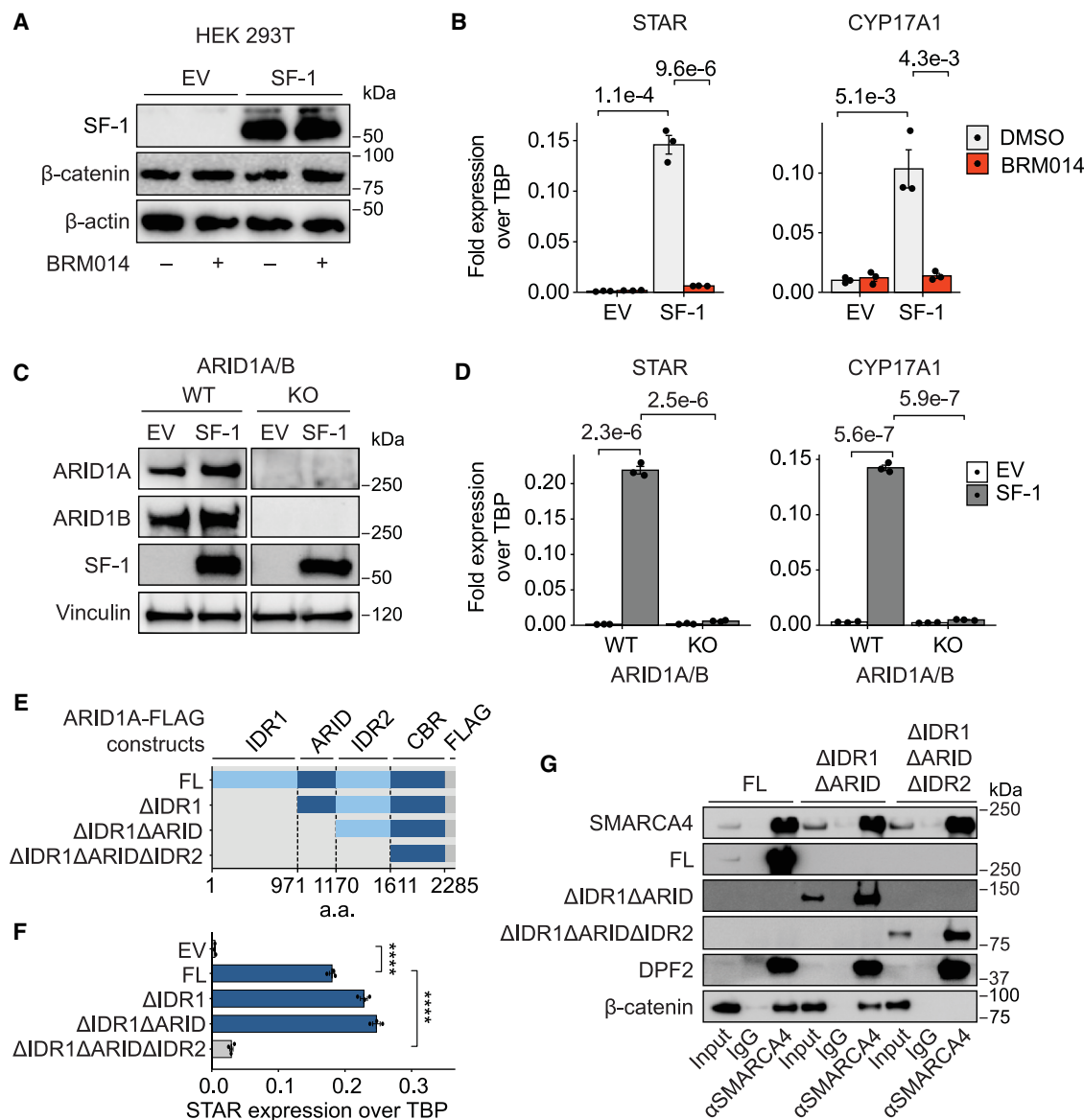


Figure 4. IDR2 of ARID1A mediates interaction of cBAF and β-catenin

(A) Western blot of ectopic expression of EV SF-1 in HEK 293T cells treated with 1 μM BRM014 or DMSO for 72 h.

(B) qPCR analysis of SF-1 targets STAR and CYP17A1 in cells shown in (A). Error bars are mean ± SEM ($n = 3$ independent replicates).

(C) Western blot of HEK 293T cells expressing wild-type (WT) or dual knockout (KO) of ARID1A and ARID1B, with or without ectopic expression of SF-1.

(D) qPCR analysis of SF-1 targets STAR and CYP17A1 in cells shown in (C). Error bars are mean ± SEM ($n = 3$ independent replicates).

(E) Design of truncation variants of ARID1A. Positions indicate amino acid (aa) number.

(F) qPCR analysis of SF-1 target STAR normalized to housekeeping gene TBP in HEK 293T cells stably expressing SF-1 with ectopic expression of constructs shown in (E). Constructs with significantly compromised SF-1 activity compared with FL ARID1A are shown in gray. Error bars are mean ± SEM ($n = 3$ independent replicates).

(G) CoIP to investigate the interaction between BAF complexes and β-catenin in cells expressing indicated ARID1A constructs ($n = 4$ independent replicates). Abbreviations are as follows: IDR, intrinsically disordered region; ARID, AT-rich interaction domain; CBR, core binding region.

**** $p < 0.0001$.

See also Figures S7 and S8.

repeats (Figures 5E, 5F, and S9B; Table S3) that were lost when repeats 4–6 were deleted (Figure 5G). Our findings demonstrate that cBAF–β-catenin association is mediated by direct contacts between ARID1A IDR2 and folded β-catenin Armadillo repeats.

3D organization of cBAF–β-catenin–TF interactions at enhancers

In contrast to exclusively IDR-mediated mechanisms, assembly of the cBAF–β-catenin–SF-1 ternary complex arises through a

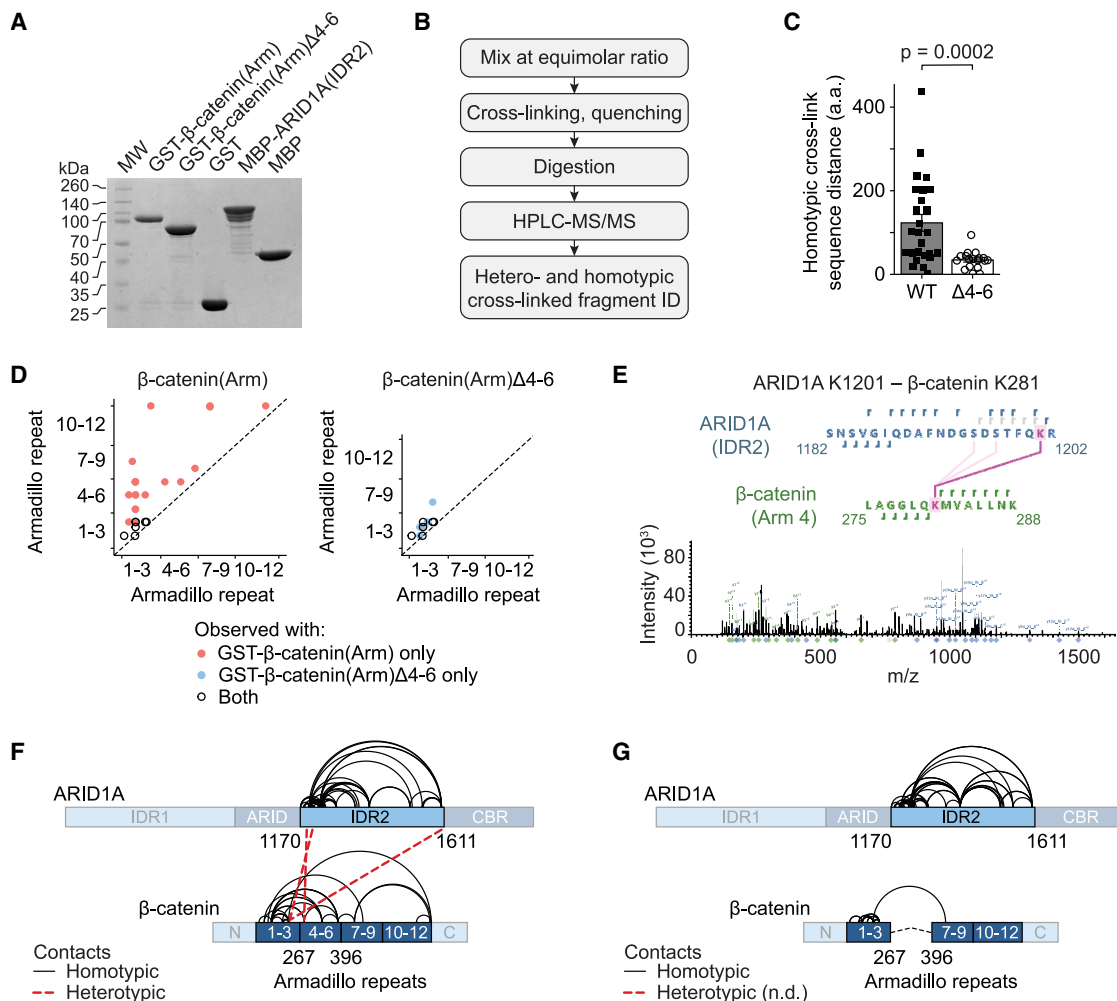


Figure 5. ARID1A IDR2 directly interacts with β-catenin folded Armadillo repeats

(A) SDS-PAGE of purified recombinant proteins used for cross-linking mass spectrometry (XL-MS).
 (B) Schematic summary of XL-MS workflow.
 (C) Distributions of sequence distance for homotypic cross-links across the β-catenin Armadillo repeat domain.
 (D) Positions of homotypic cross-links between β-catenin Armadillo repeats.
 (E) Mass spectrum corresponding to a heterotypic cross-link between ARID1A IDR2 and β-catenin (additional spectra in Figure S9).
 (F) Homotypic and heterotypic cross-links detected involving ARID1A IDR2 and intact β-catenin Armadillo repeats.
 (G) Homotypic and heterotypic cross-links involving ARID1A IDR2 and Δ4-6 β-catenin Armadillo repeats. Dashed black line indicates the deleted region within the Armadillo domain (repeats 4–6). In (F) and (G), solid black lines indicate homotypic cross-links, and dashed red lines indicate heterotypic cross-links. Abbreviations are as follows: IDR, intrinsically disordered region; ARID, AT-rich interaction domain; CBR, core binding region; Arm, Armadillo repeat; n.d., not detected.
 See also Figure S9 and Table S3.

network of interactions between IDRs and folded protein domains (Figure 6A). We therefore sought to visualize the three-dimensional structure of the cBAF-β-catenin-SF-1 ternary complex bound to chromatin. To build this structure, we integrated our XL-MS data (Figure 5) with information from cryo-EM,^{21,39} NMR,^{21,39} X-ray crystallography,⁶⁷ and mutagenesis⁶⁶ studies augmented with inferred structures of our experimentally determined interfaces using AlphaFold 3,⁶⁸ which accurately models interfaces involving IDRs.⁶⁹

The resulting dinucleosome structure (Figure 6B) revealed that ternary complexes involving cBAF, β-catenin, and TFs

like SF-1 can position cBAF within ± 1 nucleosome of the DNA containing the TF binding sequence. IDRs in this setting enable a non-rigid configuration, a property that we reason is essential for chromatin remodeling, where differences in the lengths and angles between nucleosomes are dynamic. The resulting structure, moreover, revealed that the portion of β-catenin's C-terminal IDR that binds the TAZ2 (CH3) domain of p300^{66,70} is sterically available and in an ideal position to engage p300 at the adjacent nucleosome (Figure 6B). Importantly, the size of the protein complexes and IDR lengths enable these interactions to

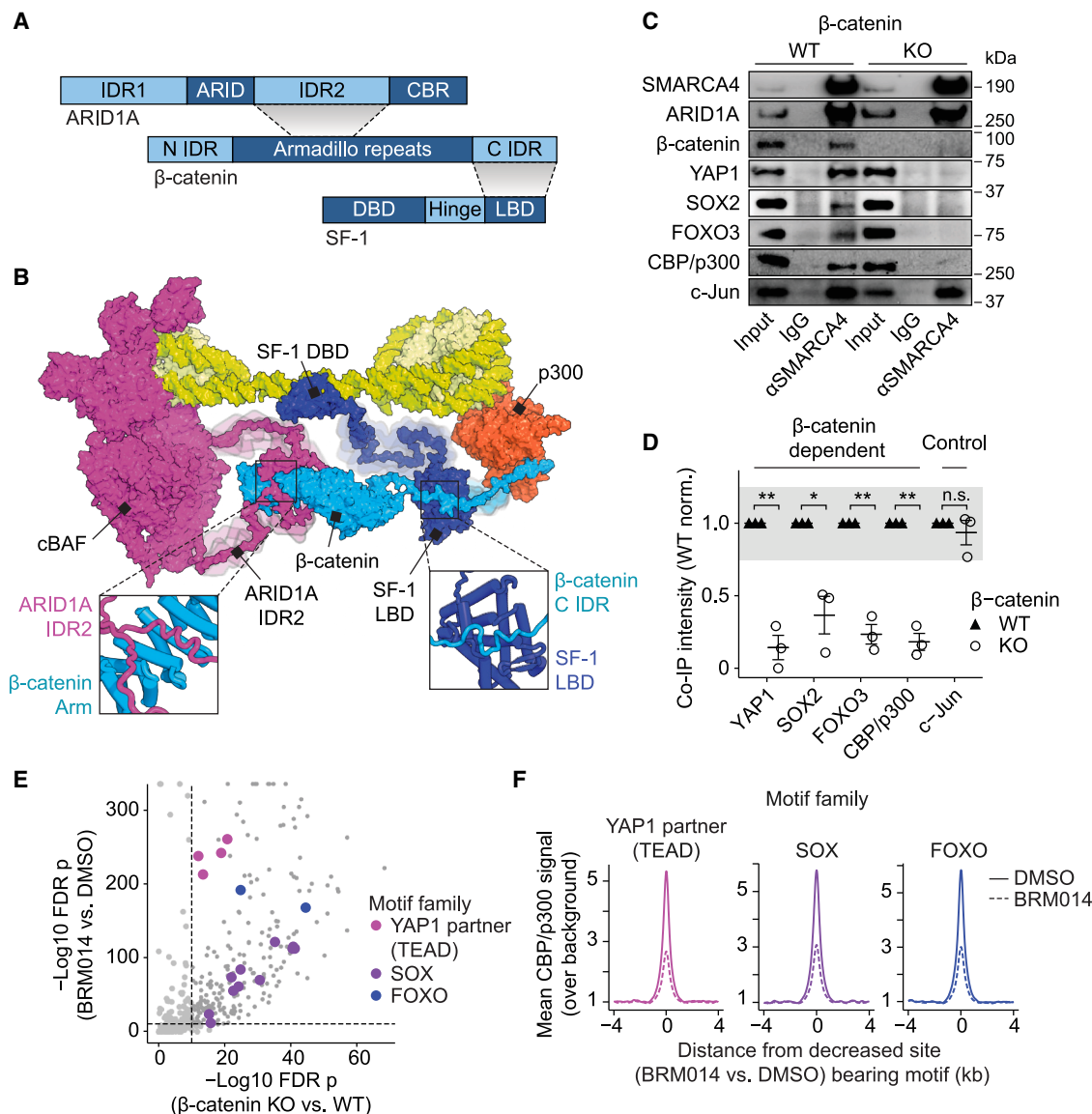


Figure 6. β-catenin acts as a general adapter for cBAF interactions with TFs and chromatin regulators

(A) Summary of interactions between cBAF, β-catenin, and SF-1. Disordered regions are shown in light blue, and structured regions are shown in dark blue. (B) Integrative three-dimensional model based on observed protein contacts between SF-1, β-catenin, and cBAF bound to a dinucleosome. Contacts are supported by experimental data, but atomic-level configurations of molecular interfaces are hypothetical. The disordered N terminus of β-catenin is not shown for clarity. (C) Co-immunoprecipitation of YAP1, SOX2, FOXO3, CBP/p300, and c-Jun control with SMARCA4 in HEK 293T cells expressing wild-type (WT) or knockout (KO) β-catenin using IgG control or anti-SMARCA4 antibody. (D) Quantification of co-immunoprecipitation densitometry in (C) ($n = 3$ independent replicates). (E) Enrichment of motifs at sites that lose DNA accessibility upon BAF inhibition or β-catenin KO. Key TF families are highlighted. (F) Metagene plots of CBP/p300 chromatin immunoprecipitation (ChIP) chromatin occupancy at sites bearing key TF motifs that lose DNA accessibility following 1 h treatment of BRM014 or DMSO.

Abbreviations are as follows: IDR, intrinsically disordered region; ARID, AT-rich interaction domain; CBR, core binding region; Arm, Armadillo repeat; DBD, DNA-binding domain; LBD, ligand-binding domain.

* $p < 0.05$, ** $p < 0.01$.

See also Table S4.

arise within the combined length span of two nucleosomes, a scale consistent with experimental DNA accessibility patterns at enhancers.⁷¹

The capacity of β-catenin to interact with different TFs^{45–48,72,73} and chromatin machinery³⁹ prompted us to speculate that β-catenin may serve as a general adapter connecting

cBAF to multiple binding partners. Excitingly, we discovered that β -catenin-interacting TFs and co-activators YAP1 (t test $p = 0.0095$, $n = 3$), SOX2 ($p = 0.039$, $n = 3$), FOXO3 ($p = 0.0076$, $n = 3$), and histone acetyltransferase CBP/p300 ($p = 0.0051$, $n = 3$) were all dependent on β -catenin for interaction with cBAF in HEK 293T cells (Figures 6C and 6D). By contrast, the AP-1 factor c-Jun, which binds directly to SMARCD1,⁷⁴ interacts independently of β -catenin ($p = 0.52$, $n = 3$; Figures 6C and 6D), serving as a control to illustrate specificity. Consistent with these findings, DNA accessibility at sites bearing motifs of TEAD (YAP1 binding partner), SOX, or FOXO families was reduced upon either β -catenin knockout or 1 h of BRM014 treatment (Figure 6E; Table S4). In agreement with our integrative three-dimensional model, CBP/p300 also displayed reduced chromatin occupancy at these sites (Figure 6F). Our results reveal that β -catenin serves as a general molecular adapter to cBAF used by many factors, physically linking the chromatin remodeling activity of cBAF to TFs and other chromatin regulators.

DISCUSSION

BAF chromatin remodeling complexes are critical regulators of gene expression, yet biochemical characterization of their interactomes remains a significant challenge. This limitation is largely attributable to the prevalence of disordered and flexible regions within these complexes, which are important for mediating protein-protein interactions. Despite substantial efforts, less than 40% of the mass of these complexes has been resolved to date.²⁸ As a result, understanding how IDRs in molecular machines such as BAF maintain their multifaceted protein interactomes represents a currently unresolved issue. Here, we discover a general IDR-mediated mechanism by which cBAF engages other transcriptional regulators by using β -catenin as a molecular adapter.

β -catenin harbors a folded domain composed of helical Armadillo repeats, a structure uniquely suited for supporting a wide range of IDR-mediated interactions.^{75–80} These properties make β -catenin an effective molecular bridge, linking cBAF complexes to SF-1, YAP1, SOX2, and FOXO3, as well as histone acetyltransferases CBP/p300. In addition to the effects on steroidogenesis we highlight here, the dependency on β -catenin for these interactions reveals a modular basis by which cBAF engages factors essential for pluripotency,⁸¹ aging,⁸² and tissue homeostasis.⁸³ Because β -catenin's interaction partners include additional transcription regulators⁴¹ that are also BAF-dependent,^{62,84,85} we anticipate that factors beyond those tested here may display similar dependency on its molecular adapter function. Our finding that structured elements mediate IDR-driven cBAF interactions further emphasizes the growing recognition that IDR selectivity is strongly influenced by interactions with folded domains.^{28,34}

Our work reveals an important chromatin regulatory axis underlying endocrine function and positions cBAF as a critical steroidogenic dependency. Our results suggest that effects on steroid production may be an important consideration for the use of BAF ATPase inhibitors currently in clinical trials as anti-cancer agents.^{86–88} In particular, production of the SF-1-dependent glucocorticoid cortisol suppresses inflammation,⁸⁹ promotes

T cell exhaustion,⁹⁰ and impairs T cell infiltration into tumors, thereby reducing the efficacy of immune checkpoint therapies.⁹¹ Elevated glucocorticoid levels in adrenal cancer indeed represent a key hurdle for the efficacy of immune therapies.^{92–96} As a result, potential changes in steroid production following BAF inhibition have the potential to exert local or systemic effects on immune microenvironments, including in tumors. By focusing on dependencies of steroid production, our work uncovers a widely used avenue for disordered protein to influence endocrine activities via the regulation of chromatin and transcription.

Limitations of the study

Our data reveal a key role for β -catenin in mediating IDR-dependent cBAF activities. Although we show this function is mediated by direct interaction, we cannot exclude the involvement of additional factors that may influence this regulatory axis. Achieving atomic-level resolution of the involved interfaces will require further structural analyses, such as cryo-EM or X-ray crystallography. Moreover, our study does not address the temporal order of events by which BAF and β -catenin associate with SF-1, CBP/p300, or other transcription regulators. Finally, although we did not detect a functional interaction between SF-1 and either PBAF or ncBAF complexes, we cannot rule out that these complexes may have direct or indirect roles in β -catenin transcriptional regulation.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, H. Courtney Hodges (chodges@bcm.edu).

Materials availability

All reagents used in this study can be found in the [key resources table](#). Plasmids generated in this study are available upon request to the [lead contact](#).

Data and code availability

- High-throughput sequencing data generated for this project have been deposited in the Gene Expression Omnibus (GEO) database with accession numbers GEO: GSE260866, GSE260867, and GSE261006. Proteomic data generated for this project have been deposited in the ProteomeXchange database with accession number PRIDE: PXD063603. Original western blot images, including independent replicates, have been deposited at Mendeley Data (<https://doi.org/10.17632/2xm4vsv6w8.1>).
- This paper does not report original code. Scripts used for genome-wide analyses are available on Zenodo⁹⁷ and at the following link: <https://github.com/hodgeslab/workflows>.
- Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

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AUTHOR CONTRIBUTIONS

Y.S.C.: formal analysis, visualization, methodology, and writing – original draft. Q.G., S.A.R., W.W., and M.L.C.: investigation and writing – review and editing. R.F., L.-M.W., C.S.R.A., A.M.L., N.P., and G.D.H.: investigation. M.Z.: resources. K.K.-V., M.E.W., M.A.H., and V.V.: investigation and resources. K.C. and H.C.H.: conceptualization, resources, formal analysis, supervision, funding acquisition, visualization, methodology, writing – original draft, project administration, and writing – review and editing.

DECLARATION OF INTERESTS

G.D.H. reports unrelated personal fees from Radionetics and Orphagen Pharmaceuticals for consultation on projects outside the scope of this work.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
SMARCA4 (IP, WB)	Santa Cruz Biotechnology	Cat# sc-374197; RRID:AB_10990135
SMARCA4 (ChIP, WB)	Proteintech	Cat# 21634-1-AP; RRID:AB_10858784
SMARCA4 (IHC)	Abcam	Cat# ab108318; RRID:AB_10889900
SMARCA2 (WB)	Abcam	Cat# ab15597; RRID:AB_443214
β -catenin (ChIP, WB)	Invitrogen	Cat# 71-2700; RRID:AB_88035
β -catenin (IHC)	Cell Signaling Technology	Cat# 8480; RRID:AB_11127855
SF-1 (ChIP, IP)	R&D Systems	Cat# PPN16650C
SF-1 (WB)	Millipore	Cat# 07-618; RRID:AB_310756
SF-1 (IHC)	Thermo Fisher	Cat# 2516-MSM7-P1
IgG (IP)	Santa Cruz Biotechnology	Cat# sc-2025; RRID:AB_737182
FLAG (co-IP, WB)	Millipore	Cat# F1804; RRID:AB_262044
ARID1A (ChIP, WB)	Cell Signaling Technology	Cat# 12354S; RRID:AB_2637010
ARID1A (IP)	Santa Cruz Biotechnology	Cat# sc-32761; RRID:AB_673396
ARID1B (WB)	Santa Cruz Biotechnology	Cat# sc-32762; RRID:AB_2060367
DPF1 (WB)	Proteintech	Cat# 21769-1-AP; RRID:AB_3085672
DPF2 (WB)	Proteintech	Cat# 12111-1-AP; RRID:AB_2095574
DPF3 (WB)	Proteintech	Cat# 68646-1-Ig; RRID:AB_3670389
PBRM1 (ChIP, WB)	Bethyl Laboratories	Cat# A301-591A; RRID:AB_1078808
BRD9 (ChIP, WB)	Abcam	Cat# ab137245
H3K4me1 (ChIP)	Abcam	Cat# ab8895; RRID:AB_306847
H3K4me3 (ChIP)	Millipore	Cat# 05-745R; RRID:AB_1587134
H3K27ac (ChIP)	Abcam	Cat# ab4729; RRID:AB_2118291
H3K27me3 (ChIP)	Active Motif	Cat# 39155; RRID:AB_2561020
CTCF (ChIP)	Millipore	Cat# 07-729; RRID:AB_441965
FOXO3 (WB)	Cell Signaling Technology	Cat# 12829; RRID:AB_2636990
SOX2 (WB)	Millipore	Cat# AB5603; RRID:AB_2286686
YAP1 (WB)	Proteintech	Cat# 13584-1-AP; RRID:AB_2218915
CBP/p300 (WB)	Proteintech	Cat# 22277-1-AP; RRID:AB_3085692
CBP/p300 (ChIP)	Abcam	Cat# ab14984; RRID:AB_301550
c-Jun (WB)	Cell Signaling Technology	Cat# 9165; RRID:AB_2130165
Vinculin (WB)	Sigma-Aldrich	Cat# V9131; RRID:AB_477629
β -actin (WB)	GeneTex	Cat# GTX629630; RRID:AB_2728646
HRP conjugated mouse anti-rabbit	Santa Cruz Biotechnology	Cat# sc-2357; RRID:AB_628497
HRP conjugated anti-mouse	Santa Cruz Biotechnology	Cat# sc-516102; RRID:AB_2687626
Chemicals, peptides, and recombinant proteins		
BRM014	This paper	N/A
FHD-286	MedChemExpress	Cat# HY-144835
AU-15330	MedChemExpress	Cat# HY-145388
cOmplete ULTRA	Roche	Cat# 5892791001
NuPAGE 4-12% bis tris polyacrylamide gel	Thermo Fisher	Cat# NP0323BOX
NuPAGE 3-8% Tris-Acetate Protein Gels	Thermo Fisher	Cat# EA03785BOX
MOPS SDS running buffer	Thermo Fisher	Cat# NP0001
Novex NuPAGE Tris-Acetate SDS running buffer	Thermo Fisher	Cat# LA0041

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
PVDF membrane	Millipore	Cat# IPFL00010
BLOTTO	Santa Cruz Biotechnology	Cat# SC2325
Clarity Western ECL	Bio-Rad	Cat# 1705061
Clarity Max ECL	Bio-Rad	Cat# 1705062
Protein G Dynabeads	Invitrogen	Cat# 10003D
DpnII	New England Biolabs	Cat# R0543S
T4 DNA ligase	New England Biolabs	Cat# M0202
Csp6I	Thermo Scientific	Cat# ER0211
NheI-HF	New England Biolabs	Cat# R3131S
AgeI-HF	New England Biolabs	Cat# R3552S
MluI-HF	New England Biolabs	Cat# R3198S
Lipofectamine 3000	Thermo Scientific	Cat# L3000001
Polyethylenimine hydrochloride	Polysciences	Cat# 247651
TRIzol	Thermo Scientific	Cat# 15596026
IPTG	Merck	Cat# I6758
Glutathione affinity resin	Merck	Cat# GE17-5132-02
Amylose resin	New England BioLabs	Cat# E8021L
Trypsin/LysC	Promega	Cat# V5073
Superdex 200 Increase 10/300 GL	Merck	Cat# GE28-9909-44
BS3 cross-linker	Thermo Scientific	Cat# 21580
C18 PepMap Neo Trap Cartridge	Thermo Scientific	Cat# 174500
Aurora Ultimate Nano reversed phase column	Ion Opticks	Cat# AUR3-25075C18
DMEM/F12	ATCC	Cat# 30-2006
Nu-serum	Corning	Cat# 355500
ITS	Corning	Cat# 354352
Penicillin/streptomycin	Gibco	Cat# 15140122
Hydrocortisone	Sigma-Aldrich	Cat# H0888
Insulin	Sigma-Aldrich	Cat# I0516
Cholera toxin	Sigma-Aldrich	Cat# C8052
Epidermal growth factor	Gibco	Cat# PHG0311L
Adenine	Sigma-Aldrich	Cat# A2786

Critical commercial assays

Testosterone ELISA	Enzo Life Sciences	Cat# ADI-900-065
DHEA ELISA	Enzo Life Sciences	Cat# ADI-900-093
Cortisol ELISA	Enzo Life Sciences	Cat# ADI-900-071
Aldosterone ELISA	Enzo Life Sciences	Cat# ADI-900-173
CYP11A1 TaqMan probe	Thermo Fisher	Hs00167984_m1
CYP17A1 TaqMan probe	Thermo Fisher	Hs01124136_m1
SCARB1 TaqMan probe	Thermo Fisher	Hs00969821_m1
STAR TaqMan probe	Thermo Fisher	Hs00264912_m1
MC2R TaqMan probe	Thermo Fisher	Hs01563483_s1
NR5A1 TaqMan probe	Thermo Fisher	Hs00610436_m1
TBP TaqMan probe	Thermo Fisher	Hs00427620_m1
TaqMan Fast Advanced Master Mix	Applied Biosystems	Cat# 4444557
Pierce Detergent Compatible Bradford Assay Kit	Thermo Fisher	Cat# 23246
Illumina Tagment DNA Enzyme and Buffer Large Kit	Illumina	Cat# 20034198
High-Fidelity 2X PCR Master Mix	New England Biolabs	Cat# M0541S
AMPure XP beads	Beckman Coulter	Cat# A63880

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
PCR beads	Nucleomag	Cat# 744100.1
MinElute PCR purification kit	Qiagen	Cat# 28004
In-Fusion	Takara Bio	Cat# 639649
Clean-Blot IP Detection Reagent	Thermo Fisher	Cat# 21230

Deposited data

ATAC-seq data	This study	GEO: GSE260866
RNA-seq data	This study	GEO: GSE261006
ChIP-seq data	This study	GEO: GSE260867
Raw western blot data	This study	Mendeley Data: https://doi.org/10.17632/2xm4vsv6w8.1
XL-MS data	This study	PRIDE: PXD063603

Experimental models: Cell lines

H295R	ATCC	Cat# CRL-2182, RRID:CVCL_0458
CU-ACC1	Kiseljak-Vassiliades et al. ⁵⁸	RRID:CVCL_RQ00
HEK 293T	ATCC	Cat# CRL-3216, RRID:CVCL_0063

Oligonucleotides

SMARCA4 shRNA #1	Dharmacon	Cat# TRCN0000015549
SMARCA4 shRNA #2	Dharmacon	Cat# TRCN0000015550
SMARCA2 shRNA #1	Dharmacon	Cat# TRCN20330
SMARCA2 shRNA #2	Dharmacon	Cat# TRCN20332
SF-1 shRNA #1	Dharmacon	Cat# TRCN0000019599
SF-1 shRNA #2	Dharmacon	Cat# TRCN0000019600
β -catenin shRNA #1	Rosenbluh et al. ⁴⁵	Addgene plasmid #42543
β -catenin shRNA #2	Rosenbluh et al. ⁴⁵	Addgene plasmid #42544
Non-targeting control	Dharmacon	Cat# RHS6848

Recombinant DNA

piggyBac transposase	C. Bashor	N/A
SF-1 ORF	Genscript	Cat# OHu26165
ARID1A ORF	NCBI	GenBank: NM_006015
β -catenin ORF	Morin et al. ⁹⁸	Addgene plasmid #16518; RRID:Addgene_16518
SF-1 Δ LBD	This study	N/A
ARID1A deletion constructs	This study	N/A
β -catenin deletion constructs	This study	N/A

Software and algorithms

Bowtie 2.4.1	Langmead and Salzberg ⁹⁹	RRID:SCR_005476
MACS 2.1.1	Zhang et al. ¹⁰⁰	RRID:SCR_013291
Diffbind	Stark and Brown ¹⁰¹	RRID:SCR_012918
DESeq2	Love et al. ¹⁰²	RRID:SCR_015687
Limma	Ritchie et al. ¹⁰³	RRID:SCR_010943
Bedtools 2.28.0	Quinlan and Hall ¹⁰⁴	RRID:SCR_006646
bwtool	Pohl and Beato ¹⁰⁵	RRID:SCR_003035
QuPath 0.2.3	Bankhead et al. ¹⁰⁶	RRID:SCR_018257
TCGAbiolinks	Colaprico et al. ¹⁰⁷	RRID:SCR_017683
GEOquery	Davis and Meltzer ¹⁰⁸	RRID:SCR_000146
ComplexHeatmap	Gu et al. ¹⁰⁹	RRID:SCR_017270
HOMER 4.11	Heinz et al. ¹¹⁰	RRID:SCR_010881

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
ChromVAR	Schep et al. ¹¹¹	RRID:SCR_026570
nnet	Venables and Ripley ¹¹²	https://cran.r-project.org/web/packages/nnet/index.html
YASARA 24.4.10	Krieger and Vriend ¹¹³	RRID:SCR_017591
Mass Spec Studio	Sarpe et al. ¹¹⁴	PMID: 27412762
pyGenomeTracks	Lopez-Delisle et al. ¹¹⁵	RRID:SCR_025312
survival	Therneau et al. ¹¹⁶	RRID:SCR_021137
survminer	Kassambara, Kosinski, Biecek	RRID:SCR_021094
pROC	Robin et al. ¹¹⁷	RRID:SCR_024286
pipe4C	Krijger et al. ¹¹⁸	https://doi.org/10.1016/j.ymeth.2019.07.014

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell lines

H295R cells were acquired (ATCC #CRL-2182) and maintained in DMEM/F12 medium (ATCC #30-2006) supplemented with 2.5% of Nu-serum (Corning #355500), 1% ITS (Corning #354352) and 1% penicillin/streptomycin (Gibco #15140122). CU-ACC1 cells⁵⁸ were maintained in F medium [3:1 (v/v) F12:DMEM], 5% FBS, 0.4 μ g/mL hydrocortisone (Sigma-Aldrich #H0888), 5 μ g/mL insulin (Sigma-Aldrich #I0516), 8.4 ng/mL cholera toxin (Sigma-Aldrich #C8052), 10 ng/mL epidermal growth factor (Gibco #PHG0311L), 24 μ g/mL adenine (Sigma-Aldrich #A2786) and 1% Pen/Strep as previously reported.⁵⁸ HEK 293T cells were acquired (ATCC #CRL-3216) and maintained in DMEM supplemented with 5% FBS.

METHOD DETAILS

Cell-line generation

CRISPR-Cas9 gene knockout

HEK 293T cells were transfected with lentiCRISPR v2 plasmids encoding mCherry, Cas9 protein and specific single guide RNAs (sgRNAs) selected from human GeCKO libraries.¹¹⁹ The following gRNA sequences were employed for gene editing in this study: ARID1A: AAT ACT CAC AGG CAA GCT GG; ARID1B: GTC CGA CCC TGG ATG CCA AT; PBRM1: GAA ACC ACT TCA TAA TAG TC; BRD9: CTT GAC GGA CAG TAC CGC AG; DPF1: GTC CGA CCC TGG ATG CCA AT; DPF2: TGG ATG GAA AAG CGA CAC CG; DPF3: CAA GTA GGC ACT CAC GCC TG. For CTNNB1 knockout, the following two gRNAs were used: GAA AAG CGG CTG TTA GTC AC and TTC CCA CTC ATA CAG GAC TT. All knockout experiments included non-targeting control (NTC) gRNA: GGG AGG TGG CTT TAG GTT TT. Following transfection, mCherry-positive cells were sorted into individual wells of a 96-well plate to obtain monoclonal cell lines. Knockout of targets was confirmed by western blot.

shRNA knockdown

To achieve stable knockdown of SMARCA4, SMARCA2, SF-1, or β -catenin, H295R cells were transduced with lentivirus expressing pLKO.1 shRNAs. Transduced cells were selected using 3 μ g/mL puromycin. All shRNAs used in this study are provided in [key resources table](#).

piggyBac transposition

An acceptor plasmid for piggyBac transposition was constructed for ectopic protein expression. ARID1A (NCBI NM_006015) or SF-1 (Genscript #OHu26165) were cloned into this plasmid with C-terminal FLAG tag. SF-1 lacking the ligand-binding domain (Δ LBD) was created by In-Fusion (Takara Bio #639649). Full-length β -catenin was obtained from pCI-neo, a gift from Bert Vogelstein⁹⁸ (Addgene plasmid #16518; <http://n2t.net/addgene:16518>; RRID:Addgene_16518), and cloned into the piggyBac expression plasmid with an HA tag through ligation. All truncation and deletion constructs of ARID1A and β -catenin were synthesized with boundaries denoted in main figures. Vectors were validated using Sanger and/or whole-plasmid sequencing.

Initial piggyBac transposition was performed using Lipofectamine 3000 (Thermo Fisher #L3000001). Cells were selected using 10 μ g/mL blasticidin. Stable integration and protein expression were confirmed by western blot.

Lentiviral transduction

HEK 293T cells were co-transfected with packaging plasmids (psPAX2, pMD2.G) and transfer plasmid using polyethylenimine hydrochloride (PEI, Polysciences #247651, MW 40,000). Culture medium was changed to DMEM with 2% FBS 16 h after transfection. Lentiviral particle-containing culture medium was harvested 72 h after transfection. The medium was filtered through a 5- μ m filter (Millipore #SLSV025LS), concentrated by ultrafiltration on a 100-kDa Amicon filter (Millipore #UFC903008), and stored at -80°C . Cells were transduced at 50% confluency using spinfection (1,000 g, 30 min, 32°C).

Inhibitors and degraders

Sourcing and synthesis

FHD-286 (MedChemExpress #HY-144835) and AU-15330 (MedChemExpress #HY-145388) were purchased commercially. BRM014 was synthesized as described earlier.^{1,2,59} For BRM014, the expected molecular weight of 318 Da was confirmed, and purity of >99% was confirmed based on LC/MS measurement.

Treatment conditions

Unless stated otherwise, 500,000 cells were seeded per well in 6-well plates. After attachment, cells were treated with 1 μ M BRM014 or equal volume of DMSO as vehicle control. Cells were then incubated for up to 72 hours. 1 μ M FHD-286 or 1 μ M AU-15330 were added in the same manner to evaluate their effects on SF-1 activity.

RT-qPCR

RNA was harvested using TRIzol reagent (Invitrogen #15596026). 1 μ g of RNA was then reverse-transcribed into cDNA using a high-capacity reverse transcription kit (Fisher #43-688-14). TaqMan qPCR reactions contained 1x TaqMan Fast Advanced Master Mix (Applied Biosystems #4444557), 1x target primers (FAM-MGB), 1x housekeeping primers (VIC-MGB), and 50 ng of cDNA in a volume of 20 μ l. TaqMan qPCR assays used in this study are provided in [key resources table](#). PCR was performed with a QuantStudio 3 Real-Time PCR system (Applied Biosystems #A28567). Quantification was internally normalized to TBP expression levels using the $\Delta\Delta$ Ct method and control comparisons were made simultaneously using equivalently passaged cells.

Western blotting

Immunoblotting

Total protein was isolated from cells using RIPA lysis buffer (50 mM Tris HCl pH 7.5, 150 mM NaCl, 1% NP-40, 0.5% sodium deoxycholate, 0.1% SDS, cOmplete ULTRA protease inhibitors (Roche #5892791001)), sonicated, and cleared by centrifugation. Protein concentration was determined using Bradford assay (Thermo Fisher #23246). Subsequently, 20 μ g of total protein per well was run on a Novex NuPAGE 4-12% bis tris polyacrylamide gel (Thermo Fisher #NP0323BOX) in MOPS SDS running buffer (Thermo Fisher #NP0001); or Nupage 3-8% Tris-Acetate Protein Gels (Thermo Fisher #EA03785BOX) in Novex NuPAGE Tris-Acetate SDS running buffer (Thermo Fisher #LA0041). Gel content was transferred to 0.45- μ m PVDF membrane (Millipore #IPFL00010) overnight.

Detection

Western blots were blocked in 5% BLOTTO (Santa Cruz Biotechnology #SC2325) dissolved in Tris-buffered saline with 0.1% Tween 20 Detergent (TBS-T), incubated in primary antibody overnight at 4°C, washed 3 times in TBS-T, and probed with secondary antibody for 1-2 hours at room temperature (RT). Antibodies are provided in [key resources table](#). Clean-Blot IP Detection Reagent (Thermo Fisher #21230, 1:1,000 dilution) was used for detecting co-IP samples to avoid contamination from denatured antibodies. Detection of HRP-conjugated antibodies was performed using Clarity Western or Clarity Max ECL substrates (Bio-Rad). Visualization was performed on a Bio-Rad ChemiDoc MP imager with densitometry analysis performed using Image Lab software (6.1).

Co-immunoprecipitation

To assess protein interactions via co-immunoprecipitation (coIP), H295R or HEK 293T cells were first lysed with IP buffer (25 mM Tris HCl pH 8.0, 150 mM NaCl, 1% NP-40, 1 mM EDTA) for 10 minutes on ice. Lysed samples were sonicated for 6 cycles of 10 seconds on and 30 seconds off at 4°C and cleared by centrifugation. 500 μ g of whole-cell lysate was pre-incubated with protein G dynabeads (Invitrogen #10003D) for 30 minutes and removed using a magnetic stand. Lysates were then incubated with 5 μ g antibodies overnight. The targeted proteins and antibodies were incubated with protein G beads for 1 hour followed by magnetic separation. Protein was eluted by adding 1x sample buffer (Thermo Fisher #NP0007) and boiled at 95°C for 5 minutes. Antibodies are provided in [key resources table](#). Before running western blot, eluted samples were supplemented with 2.5% β -mercaptoethanol and incubated at 95°C for 5 minutes, except for SOX2, where it was omitted to avoid signal overlap with antibodies.

Chromatin immunoprecipitation and sequencing

Library preparation

Chromatin immunoprecipitation (ChIP) was performed as previously described.³ Briefly, 10⁷ H295R or 8x10⁶ HEK 293T cells were seeded into 15-cm dishes. After attachment, cells were treated with 1 μ M BRM014 or equal volume of DMSO for 1 hour. ChIP libraries were prepared from a single-cell suspension fixed for 10 minutes in 1% formaldehyde (histone markers, CTCF, SF-1), or dual-fixed for 30 minutes in 2 mM disuccinimidyl glutarate followed by a 10-minute incubation in 1% formaldehyde (BAF subunits, β -catenin, CBP/p300). Excess formaldehyde was quenched by the addition of glycine to a final concentration of 125 mM. Fixed cells were washed, pelleted, and snap-frozen using liquid nitrogen. Antibodies used are provided in [key resources table](#).

Analysis

Sequencing reads were mapped to the hg38 human reference genome using Bowtie (2.4.1).⁹⁹ Duplicate fragments and reads with mapping quality <10 were discarded. Peak calling was performed by MACS (2.1.1),¹⁰⁰ Diffbind¹⁰¹ and DESeq2¹⁰² were used for differential peak calling and for estimating size factor. Differential peak calls were made by requiring fold changes of >2 fold in either direction and FDR-adjusted p-values <0.05. Overlap of peaks was assessed using bedtools (2.28.0).¹⁰⁴ Calculation of mean densities

and preparation of genome-wide heatmaps was performed using bwtool.¹⁰⁵ Genome annotation was done with ChromHMM⁶³ using genome-wide profiles of CTCF, H3K4me1, H3K4me3, H3K27ac, H3K4me3, and H3K27me3.

RNA sequencing

Library preparation

500,000 H295R or 300,000 CU-ACC1 cells were seeded per well in 6-well plates. After attachment, cells were treated with 1 μ M BRM014 or equal volume of DMSO for 72 hours. Cells were harvested using TRIzol reagent. mRNA was isolated by polyA-enrichment, fragmented, and reverse transcribed into cDNA. Paired-end sequencing was performed on an Illumina NovaSeq 6000.

RNA-seq data analysis

As previously described,³ reads were aligned to the hg38 human reference genome utilizing HISAT2.¹²⁰ Reads with mapping quality below 10 were excluded from further analysis. Reads mapping to specific genes were quantified using HTSeq.¹²¹ Differential gene expression analysis was processed with DESeq2,¹⁰² using default parameters. Genes with expression fold changes exceeding 1.5-fold in either direction, coupled with FDR-adjusted *p* values less than 0.05, were deemed differentially expressed.

Gene set enrichment analysis

Gene set enrichment analysis was performed using the R package fgsea (1.27.0).¹²² The following gene sets were used to generate enrichment plots: KEGG_STEROID_BIOSYNTHESIS, WEST_ADRENOCORTICAL_TUMOR_UP, PID_BETA_CATENIN_NUC_PATHWAY, and SF-1 targets (derived from published annotations.⁵²)

ATAC sequencing

Library preparation

500,000 H295R or HEK 293T cells were seeded per well in 6-well plates. After attachment, cells were treated with 1 μ M BRM014 or an equal volume of DMSO for 1 hour. As previously described,^{1,2} nuclei were isolated by resuspending cells in 50 μ l lysis buffer (comprising 0.1% Tween-20, 0.1% NP-40, 0.01% digitonin, and RSB buffer with 10 mM Tris-HCl pH 7.5, 10 mM NaCl, 3 mM MgCl₂) and incubated on ice for 3 min. Cells were then washed with RSB buffer supplemented with 0.1% Tween-20 to remove residual lysis buffer. Nuclei were centrifuged at 500 g for 10 min to pellet.

For transposition, nuclear pellets were resuspended in 50 μ l of Transposition mix (2.5 μ l Tagment DNA enzyme (Illumina #20034198), 25 μ l of Tagment DNA buffer (Illumina #20034198), 0.1% Tween-20 and 0.01% Digitonin) and incubated at 37°C for 30 min. Post-incubation, transposed DNA was purified using the MinElute PCR purification kit (Qiagen #28004). Libraries were amplified using barcoded Nextera primers (Illumina) and 2x NEBNext High-Fidelity PCR Master Mix (NEB #M0541S). Finally, libraries were purified and size-selected with AMPure XP beads (Beckman Coulter #A63880), selecting for fragment sizes ranging from ~100 to 1,000 bp.

Analysis

As described previously,^{1,2} ATAC sequencing (ATAC-seq) reads were mapped to the hg38 human reference genome using Bowtie (2.4.1).⁹⁹ Duplicate fragments and reads with mapping quality <10 were discarded. Peak calling was performed by MACS (2.1.1).¹⁰⁰ DESeq2 was used for differential peak calling, with size factors determined using the top 10% of accessible sites. Differential peak calls were made by requiring fold changes of >1.5-fold in either direction and FDR-adjusted *p* values <0.10. TF motif enrichment and significance were measured using HOMER (4.11)¹¹⁰ and ChromVAR.¹¹¹

4C library preparation, sequencing, and analysis

Identification of enhancers for NR5A1 and CTNNB1

We used 4C to experimentally identify enhancers that interact with the NR5A1 promoter. Enhancers for CTNNB1 were identified by the overlap of ChIP-seq histone modification data with adrenal promoter capture Hi-C (pcHi-C), as previously described.^{48,123}

Cross-linking and nuclei isolation

As previously described,¹¹⁸ ~10 million H295R cells were cross-linked using 1% formaldehyde at RT for 10 minutes. The reaction was quenched with 0.13M glycine. Nuclei were then isolated using a lysis buffer composed of 50 mM Tris-HCl (pH 7.5), 0.5% NP-40, 1% TX-100, 150 mM NaCl, 5 mM EDTA, and 1x protease inhibitors. Isolated nuclei were resuspended in 1.2x NEBuffer supplemented with 0.3% SDS and 2.5% Triton.

DNA digestion and ligation

Resuspended nuclei were digested with 100 U of DpnII (NEB #R0543S) at 37 °C for 3 hours, followed by an additional overnight incubation with another 100 U of DpnII. DpnII was inactivated by incubating at 65°C for 20 minutes. The digested samples were then ligated in 7 mL of ligation buffer using 3333 U of T4 DNA ligase (NEB #M0202) at 16°C overnight. De-crosslinking was achieved by adding 30 μ l of Proteinase K (10 mg/ml) and incubating overnight at 65°C. DNA was purified using PCR beads (Nucleomag #744100.1). A subsequent digestion step was performed using 50 U of Csp6I (Thermo Scientific #ER0211) at 37°C overnight. For the second ligation, 25 μ g of template DNA was mixed with 500 μ l of 10x ligation buffer (660 mM Tris-HCl, 50 mM MgCl₂, 50 mM DTT, 10 mM ATP), 3333 U of T4 Ligase, and adjusted to a total volume of 5 ml with milli-Q water. The mixture was then incubated overnight at 16°C and the 4C templates were purified using PCR beads.

PCR amplification

Inverse PCR was used to amplify fragments ligated to the viewpoint. Primers were specifically designed using 4C-primer designer.¹²⁴ One reading primer and two non-reading primers were designed as follows: reading primer: 5'-TAC ACG ACG CTC TTC CGA TCT TCC CAC GGT GAC TGG ATC-3'; Non-reading primer 1: 5'-ACT GGA GTT CAG ACG TGT GCT CTT CCG ATC TTG TGG TGG GGA GCT GTT CA-3'; Non-reading primer 2: 5'-ACT GGA GTT CAG ACG TGT GCT CTT CCG ATC TGG AGC TGT TCA GAG GCG G-3'. For each non-reading primer, four 50- μ l PCR reactions were set up using the Expand Long Template PCR system (Roche #11681834001), with each containing 200 ng of 4C template. The PCR program settings were: 94°C for 2 min, 16 cycles of 94°C for 15 s, 55°C for 1 min, 68°C for 3 min, followed by a final elongation at 68°C for 5 min. The 4 PCR reactions were pooled together, and the DNA was purified with AMPure XP beads (Beckman Coulter #A63880). A second round of PCR was conducted using a universal primer and index primers for Illumina sequencing, following the PCR conditions: 94°C for 2 min, 20 cycles of 94°C for 10 s, 60°C for 1 min, 68°C for 3 min, and a final step at 68°C for 5 min.

Sequencing

Final libraries were sequenced on the Illumina MiSeq platform using the MiSeq Reagent Kit v3 (Illumina #MS-102-3001) to obtain 75 bp single-end reads.

Analysis

Sequencing reads from the two non-reading primers were combined. The combined fastq files were then processed by pipe4C¹¹⁸ and mapped to hg38 for peak calling. Genome browser tracks were generated using pyGenomeTracks.¹¹⁵

Hormone profiling

Conditioned medium

500,000 H295R or 300,000 CU-ACC1 cells were seeded into each well of a 6-well plate. After overnight seeding, cells were treated with 1 μ M of BRM014 or equal volume of DMSO. Conditioned media was collected after 72 hours of BRM014 or DMSO treatment.

ELISA

Testosterone, DHEA, cortisol, and aldosterone were measured from conditioned media by the following ELISA kits from Enzo Life Sciences: ADI-900-065, ADI-900-093, ADI-900-071, and ADI-900-173 according to the manufacturer's instructions.

Liquid chromatography/mass spectrometry

Deoxycorticosterone, corticosterone, and 11-deoxycortisol were measured via LC/MS. Steroids from conditioned media were extracted using organic and aqueous solvents [water: methanol: acetonitrile (1:1.5:2.25)] with isotopically labeled internal standard. Following extraction from conditioned media, samples were vortexed for 30 s and allowed to settle for 5 min at RT. Methyl tert-butyl ether (MTBE) was added, and the upper layer was transferred into a glass vial. The samples were dried at a low boiling point using a speed vacuum and reconstituted with 100 μ L of methanol: water (50/50 v/v). Separation and analysis were performed using the Agilent 1290 Infinity Liquid Chromatography (LC) system integrated with a 6495 Triple Quadrupole mass spectrometer (Agilent Technologies, Santa Clara, CA). Separation was achieved with a LC column (RESTEK Pinnacle DB Biphenyl 1.9 μ m [50 $\times$ 2.1 mm]) and mobile phase A and B consisting of 0.25 mM ammonium fluoride in water and 0.25 mM ammonium fluoride in methanol respectively. Data were acquired via multiple reaction monitoring (MRM) with positive electrospray ionization (ESI) mode using a 6495 Triple Quadrupole mass spectrometry through Agilent Mass Hunter Software (Agilent Technologies). Identified peaks and retention time were manually reviewed. The relative peak area was log2 transformed, followed by internal standard normalization for each method.

Protein purification

Recombinant expression plasmids were synthesized, then transformed into BL21(DE3) *E. coli*, and cultured in LB medium supplemented with ampicillin at 37°C. Protein expression was induced with 1 mM IPTG (Merck #16758-10G) for GST-tagged constructs and MBP, and with 0.25 mM IPTG for MBP-ARID1A(IDR2). GST-tagged proteins and MBP were expressed for 4 h at 37°C, while MBP-ARID1A(IDR2) was expressed overnight at 18°C. Cells were harvested by centrifugation and lysed by sonication (30 s on, 2 min off per cycle, 16 cycles, on ice) in lysis buffer containing 20 mM HEPES pH 8.0, 500 mM NaCl, and 1 mM DTT. Clarified lysates were applied to glutathione affinity resin (Merck #GE17-5132-02) or amylose resin (New England Biolabs #E8021L) and purified using manufacturer's instructions. Eluted fractions were dialyzed into gel filtration buffer (20 mM HEPES pH 8.0, 150 mM NaCl, 1 mM TCEP) and further purified on a size-exclusion column (Merck #GE28-9909-44).

Cross-linking mass spectrometry

Sample preparation

The following pairs of protein constructs were mixed in equimolar ratios: GST- β -catenin(Arm) with MBP-ARID1A(IDR2), GST- β -catenin(Arm) Δ 4-6 with MBP-ARID1A(IDR2), GST with MBP-ARID1A(IDR2), and MBP with GST. Following mixing in equimolar ratios, 5 μ g per replicate ($n=2$) of each complex was treated with BS3 (bis(sulfosuccinimidyl)suberate) cross-linker (ThermoFisher #21580) in 2x molar excess over lysines present in each sample. BS3 cross-linker was dissolved in water and added in two increments with each reaction taking 30 minutes at RT, followed by quenching by addition of 1 M Tris pH 8.8 to 50-fold molar excess over cross-linker. The resulting mixture was further incubated for 20 min at RT.

In-solution digestion

TEAB buffer (0.1 M triethyl ammonium bicarbonate, pH 8.5) in final concentration 85 mM was added to each sample to adjust pH followed by addition of sodium deoxycholate (2% final concentration). TCEP was dissolved in 100 mM TEAB and added in final concentration 10 mM. After addition of chloracetamide (25 mM final concentration), samples were incubated 30 min at 37°C in the dark. Samples were digested with Trypsin/LysC (Promega #V5073, 1:10 w/w ratio to protein) overnight at 37°C. After incubation, samples were acidified with TFA to pH 2 and the precipitated sodium deoxycholate was removed by ethyl acetate extraction. Samples were dried on speed-vac and desalted by using C18 stage tips. All samples were dried again and dissolved in 25 μ l of 0.1% TFA.

Liquid chromatography/mass spectrometry

Peptides were analyzed using an UltiMate 3000 RSLC nano HPLC system integrated with an Orbitrap Fusion Lumos Tribrid mass spectrometer (Thermo Fisher Scientific). Samples were loaded onto a trap column (Thermo Scientific #174500) for 2.000 min at 25.000 μ l/min. Loading buffer was composed of water, 2% acetonitrile and 0.1% trifluoroacetic acid. Nano reversed phase column (Ion Opticks #AUR3-25075C18) was used for LC/MS analysis. Peptides were eluted by mobile phase A (0.1 % formic acid in water, v/v) and mobile phase B (0.1 % formic acid in ACN, v/v) over 140-min (for IDR2- and Armadillo-containing proteins), or 60-min (for MBP- and GST-only control samples) gradient profiles.

Eluted peptides were immediately ionized (1600 V, 275°C) and scanned using the Orbitrap analyzer between 350–1400 m/z at a resolution (R) of 120,000, normalized AGC target 250% with automatic maximum injection time mode, normalized AGC target set to 500%, R=30,000 and dynamic maximum injection time mode. Data-dependent acquisition mode was used. Ions were isolated in a 1.6-Th window and fragmented by higher-energy collisional dissociation (HDC) with a normalized collision energy of 30%. Cycle time was set to either 1.5 s (60-min gradient) or 3 s (140-min gradient). Precursor ions with charge states ranging from +2 to +8 were selected for fragmentation. Dynamic exclusion was enabled after 2 scans (140-min gradient) or 3 scans (60-min gradient) with exclusion duration of 15 s and an exclusion window of 10 ppm.

Data analysis

Mass Spec Studio (2.4.0.35615, Trajan Scientific and Medical)¹¹⁴ was used to identify cross-linked peptides. MS mass tolerance was set to 5 ppm, MS2 mass tolerance 10 ppm, and results were filtered for cross-linked peptide level with FDR-corrected $p < 0.05$. Cysteine carbamidomethylation was used as a static modification and methionine oxidation was employed as a variable modification. Peptides with a minimum length of 5 amino acids and maximum length of 40 were included in the search. Trypsin K/R was selected as enzyme. The first linkable site for BS3 cross-linker was defined as the amino group of lysine and the N-terminal amine group whereas the second linkable site included lysine, serine, threonine, and tyrosine. All acquired cross-links were validated manually. Cross-links from both replicates were merged and filtered for peptides occurring outside the GST or MBP tags. Protein sequences of the recombinant constructs were used to identify cross-linked peptides. Residue positions reported in [Table S3](#) were converted to the corresponding positions in the canonical UniProt sequences for human ARID1A (O14497) and CTNNB1 (P35222).

Analysis of patient samples

Tissue microarray immunohistochemistry

Human tumor samples and de-identified clinical data were obtained under protocols approved by the IRB of MD Anderson Cancer Center. Slides were sectioned at 4 microns, hydrated with water, and antigen retrieved at pH 6.0. TMA slides were stained with primary antibodies provided in [key resources table](#), followed by anti-rabbit or anti-mouse HRP. Slides were then visualized with diaminobenzidine, counterstained with hematoxylin, dehydrated, and mounted with coverslips. H-score quantification of intensities was performed using QuPath (0.2.3).¹⁰⁶ Z-scaled H-scores were used for unsupervised hierarchical clustering using ComplexHeatmap¹⁰⁹ to classify specimens into SES-high, -intermediate, and -low.

Analysis of TCGA data

Public mRNA datasets and metadata were downloaded using R packages TCGAbiolinks (2.16.1)¹⁰⁷ and GEOquery (2.56.0).¹⁰⁸ SMARCA4-high was defined based on maximum difference in log-rank test statistic. SMARCA2-high and SMARCA2-low were determined by ranking tumors based on expression levels and matching the group sizes to those of SMARCA4-high and SMARCA4-low. Hazard ratio was calculated using survival.¹¹⁶ Kaplan-Meier curves were plotted with survminer (0.4.9). Differential analyses involving SMARCA4 expression or steroid phenotype were performed using limma.¹⁰³ HSP and LSP were previously annotated.^{56,125} Log2 counts per million (CPM) of SF-1, β -catenin, SMARCA4 of each tumor sample were Z-scaled, and were then used to classify specimens into SES-high, -intermediate, and -low based on unsupervised hierarchical clustering using ComplexHeatmap. To evaluate the efficiency of predicting HSP, specimens were divided randomly in equal portion into training and testing datasets. Multinomial log-linear models were developed and optimized using the training dataset with nnet (7.3-19).¹¹² Model performance was subsequently evaluated on the testing dataset. ROC analysis was performed using pROC (1.18.0).¹¹⁷

RNA expression in human tissues

The normalized transcripts per million (TPM) of NR5A1 expression in human tissues were acquired from The Human Protein Atlas consensus data.

Structural model of enhancer activation

The three-dimensional structural model was generated using YASARA (24.4.10)^{3,113} informed by published structures as well as coIP and XL-MS results. The structures of BAF (PDB 6LTJ)²¹ and p300 (PDB 7W9V)¹²⁶ bound to nucleosomes were used as

starting templates. The dinucleosome was modeled with linear 46-bp DNA containing the SF-1 motif obtained from HOMER (CAAGGTCACG). The structure of the zinc finger DNA-binding domain (DBD) of SF-1 (aa 1-118) in complex with DNA containing the SF-1 motif was predicted using AlphaFold3 (AF3). The β -catenin C-terminal domain (CTD, aa 665-781) was modeled via AF3 in complex with a region on the SF-1 ligand-binding domain (LBD, aa 221-243) previously identified as critical for this interaction by mutagenesis,⁶⁶ incorporating modifications reported on PhosphoSitePlus.¹²⁷ The final 30 amino acids of the β -catenin C-terminal IDR are sufficient for interaction with the p300 TAZ2 domain by protein NMR³⁹ while also being dispensable for SF-1 binding.⁶⁶ Based on these data, we generated a hypothetical interface between the p300 TAZ2 domain and the final 30 amino acids of β -catenin CTD using AlphaFold3 (AF3). Subsequently, the disordered CTD was linked to the structure of β -catenin Armadillo repeats (aa 134-664, PDB 1JPW).⁶⁷ We used AF3 to model the interaction of ARID1A IDR2 (aa 1210-1230 and 1597-1610) and β -catenin Armadillo repeats 1-5 (aa 161-350), which we detected to have intermolecular cross-links. The individual complexes were aligned and linked using YASARA's build tool, then subjected to energy minimization.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed in R (3.6.1) as two-sided tests. Unless otherwise specified, statistical analyses between groups are done by t test. For multiple-comparison tests, *p* values were adjusted using the Benjamini–Hochberg FDR correction procedure. All *n* values indicate independent biological replicates, genome-wide studies used *n* = 2 per condition in agreement with recommendations of the ENCODE Consortium.